

NIKITA CHAURASIA

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [GeeksforGeeks](#)

TECHNICAL SKILLS

Programming: Python, C++, SQL

CS Subjects: Data Structures & Algorithms (DSA), Operating Systems, Database Management Systems (DBMS), Computer Networks, Object Oriented Programming (OOP)

Problem-Solving: Algorithm Design, Complexity Analysis, Debugging & Optimization, System Design

Analytical Skills: Logical Reasoning, Pattern Recognition, Data Interpretation

SUMMARY

Results-driven **Software Engineer** with a strong foundation in **Electronics, AI/ML, and Data Structures and Algorithms**. Skilled in building **secure, scalable applications** and applying **machine learning** to real-world data pipelines, with a passion for **system design, efficient algorithms**, and software that performs reliably under load.

PROJECTS

SENTINEL — AI Prompt Security Gateway

June 2026

Python, Chrome Extension (Manifest V3), React, Chart.js, FastAPI, JWT

- Built a **Manifest V3 Chrome extension** that intercepts prompts on ChatGPT, Claude, and Gemini, scanning for **API keys, credentials, and PII** before submission with **zero network calls in the detection path**.
- Implemented **5 algorithms from scratch** — Aho-Corasick, Shannon entropy, Trie, Max-Heap, and Merge-Intervals — detecting **24 secret patterns** plus base64-encoded payloads via a decode-and-rescan layer.
- Shipped a React + Chart.js dashboard backed by a **FastAPI + JWT** service, 3 tunable sensitivity levels, and **104 tests** covering **ReDoS safety** and a false-positive corpus.

Live Demo: nikitayk.github.io/SENTINEL | GitHub: github.com/nikitayk/SENTINEL

ADPULSE — Real-Time Demand-Side Platform

June 2026

Python, LightGBM, Flask-SocketIO, Three.js

- Built a **full-stack Demand-Side Platform (DSP)** that participates in real-time ad auctions, prices each bid **in-memory** at **~5 ms**, and sustains a live feed of **7 bids/sec**.
- Trained **2 LightGBM models (CTR + CVR)** on IPinYou competition logs with **12 engineered features**; bid pricing scales up to **11x the base price** with a budget pacer and a **hard cap of 300**.
- Engineered a **generator-based pipeline** that streams multi-GB logs in **O(1) memory**; a **Flask-SocketIO + Three.js 3D dashboard** renders live KPIs, validated across a **Python 3.9/3.11 CI matrix** with **46 tests**.

Deployed Link: adpulse-457r.onrender.com | GitHub: github.com/nikitayk/ADPULSE

EDUCATION

B.Tech in ECAM (ECE + AI/ML)

2023 – 2027

Netaji Subhas University of Technology — **7.03 CGPA**

Senior Secondary School (CBSE Class XII)

2023

Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi — **83%**

Secondary School (CBSE Class X)

2021

Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi — **95.5%**

CERTIFICATIONS

- [AI For Everyone — DeepLearning.AI](#)
- [Supervised Machine Learning \(Regression and Classification\) — Stanford Online](#)
- [Advanced Learning Algorithms — DeepLearning.AI](#)

ACHIEVEMENTS / EXTRACURRICULAR

- Solved **400+** Data Structures and Algorithms problems on **LeetCode** and **GeeksforGeeks**, demonstrating consistent problem-solving rigor.
- Ranked within the **top 4 percentile** among **2,500+ teams** at DTU Execute 4.0 Hackathon for technical execution under time constraints.
- Secured a **Top 20 finish** at Hackmania (**Microsoft Azure**) and The Elites (**Microsoft Gurugram**), competing on cloud and enterprise-track challenges.